

sources_1\new\LED_DISPLAY.vhd

```

1  library IEEE;
2  use IEEE.STD_LOGIC_1164.ALL;
3  use IEEE.NUMERIC_STD.ALL;
4  --*****
5  --*
6  --* Name: LED_DISPLAY
7  --* Designer: Matthew Mwangi
8  --*
9  --*      This component is in charge of mapping the ball's position to the on
10 --*      board leds
11 --*
12 --*****
13
14 entity LED_DISPLAY is
15     Port (
16         pos:    in std_logic_vector(3 downto 0); --Ball's current position
17         clock: in std_logic;
18         reset: in std_logic;
19
20         leds: out std_logic_vector(15 downto 0)
21     );
22 end LED_DISPLAY;
23
24 architecture LED_DISPLAY_ARCH of LED_DISPLAY is
25     constant ACTIVE: std_logic := '1';
26 begin
27
28     DISPLAY:process(reset, clock, pos)
29     begin
30         if reset = ACTIVE then
31             leds <= (others=>'0');
32         elsif rising_edge(clock) then
33             leds <= (others=>'0');
34             leds(to_integer(unsigned(pos))) <= ACTIVE; --maping ball's position to led position
35         end if;
36     end process DISPLAY;
37
38 end LED_DISPLAY_ARCH;

```

